

Abstract

- Rapid growth of **Electric Vehicles (EVs)** increases stress on distribution grids.
- Uncontrolled EV charging may increase **peak electricity demand**.
- **Vehicle-to-Grid (V2G)** allows EV batteries to support the grid by supplying energy.
- Uncontrolled V2G may cause:
 - battery over-discharge
 - reduced driving range.
- This work studies a **bidirectional EV-grid system (G2V + V2G)**.
- A control strategy based on **battery SOC and electricity price signals** is implemented.
- The controller enables:
 - charging during **low electricity price**
 - V2G during **high electricity price**.
- Simulation demonstrates **safe SOC operation and controlled bidirectional power flow**.

Motivation and Introduction

Motivation

- EV adoption is rapidly increasing worldwide.
- High EV charging demand may create **grid congestion and peak load issues**.
- EV batteries represent **large distributed energy storage capacity**.
- V2G technology enables **bidirectional power exchange between EVs and the grid**.

Problem

- Uncontrolled charging increases **grid peak demand**.
- Uncontrolled V2G may lead to:
 - battery degradation
 - insufficient energy for driving.

Objective

- Develop a **bidirectional EV-grid energy management model**.
- Implement **SOC-aware V2G control**.
- Incorporate **electricity price signals** for intelligent charging decisions.
- Analyze system performance through **simulation**.

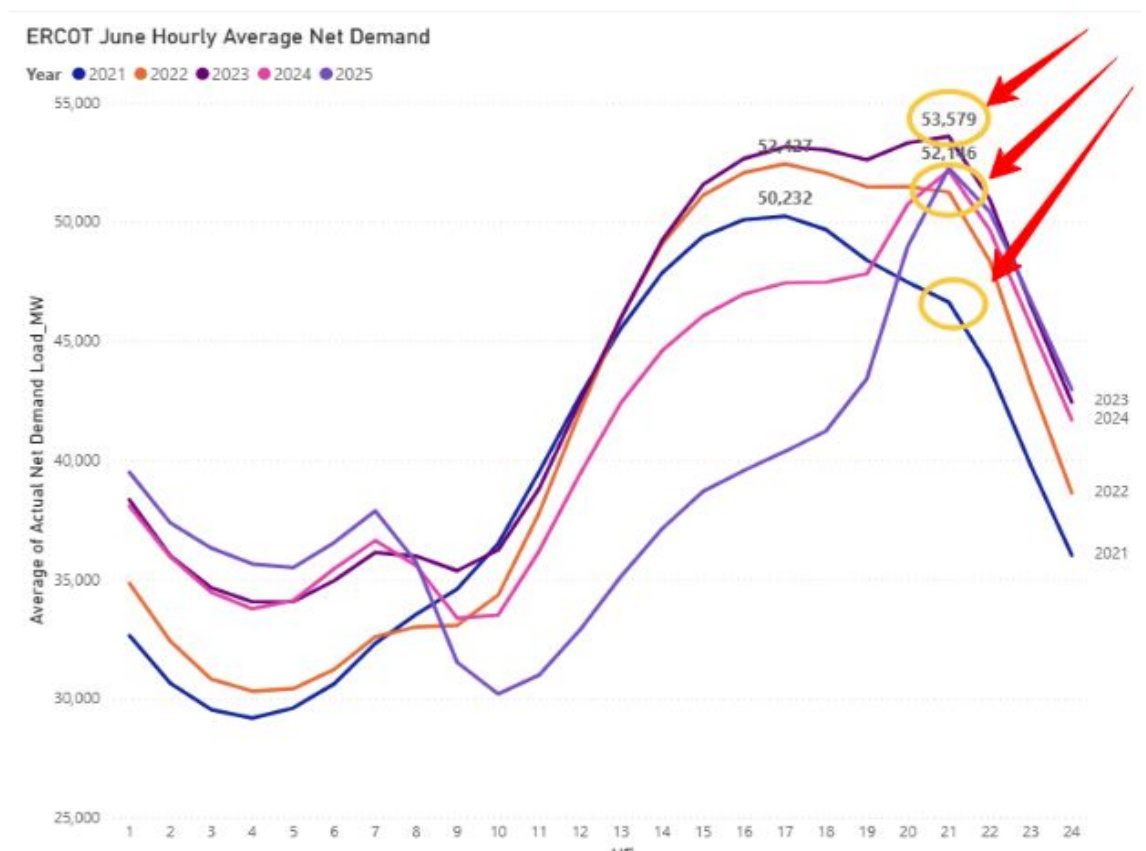


Fig. 1: Hourly Average Net Demand [5]

Literature Review

Existing Work

- Bidirectional EV chargers enable **Grid-to-Vehicle (G2V) charging and Vehicle-to-Grid (V2G) power support** for smart grids [1].
- Prior work proposes **coordinated EV charging/discharging and energy management strategies** to reduce peak demand and improve grid stability [2],[4].
- These approaches typically use **optimization-based scheduling and system-level energy management algorithms** [3],[4].

Research Gap

- Only few studies combine detailed bidirectional charger modeling with SOC and price-aware EV control in grid-connected simulations [1],[2].

Contribution of This Works

- Simulation of a **complete EV-grid interaction system** with bidirectional AC-DC and DC-DC converters [2],[6].

Methodology

Overall System Architecture

- A **bidirectional EV-grid system** is modeled in MATLAB/Simulink.

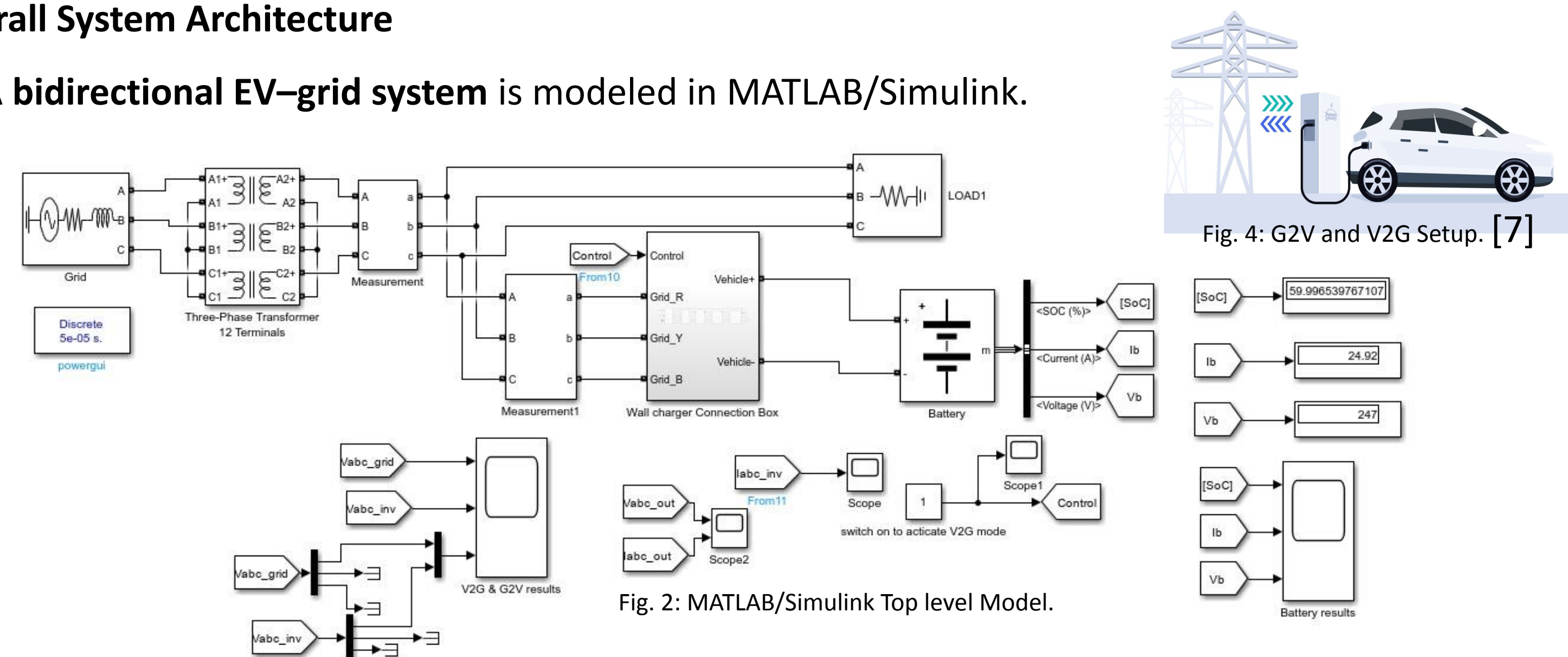


Fig. 2: MATLAB/Simulink Top level Model.

Bidirectional Charger Design

The EV charger consists of two power-electronics stages:

- **AC-DC Converter**
 - Interfaces the EV with the three-phase grid.
 - Enables **bidirectional power flow**.
- **DC-DC Converter**
 - Implemented as a **bidirectional buck-boost converter**.
 - Controls battery charging and discharging current.

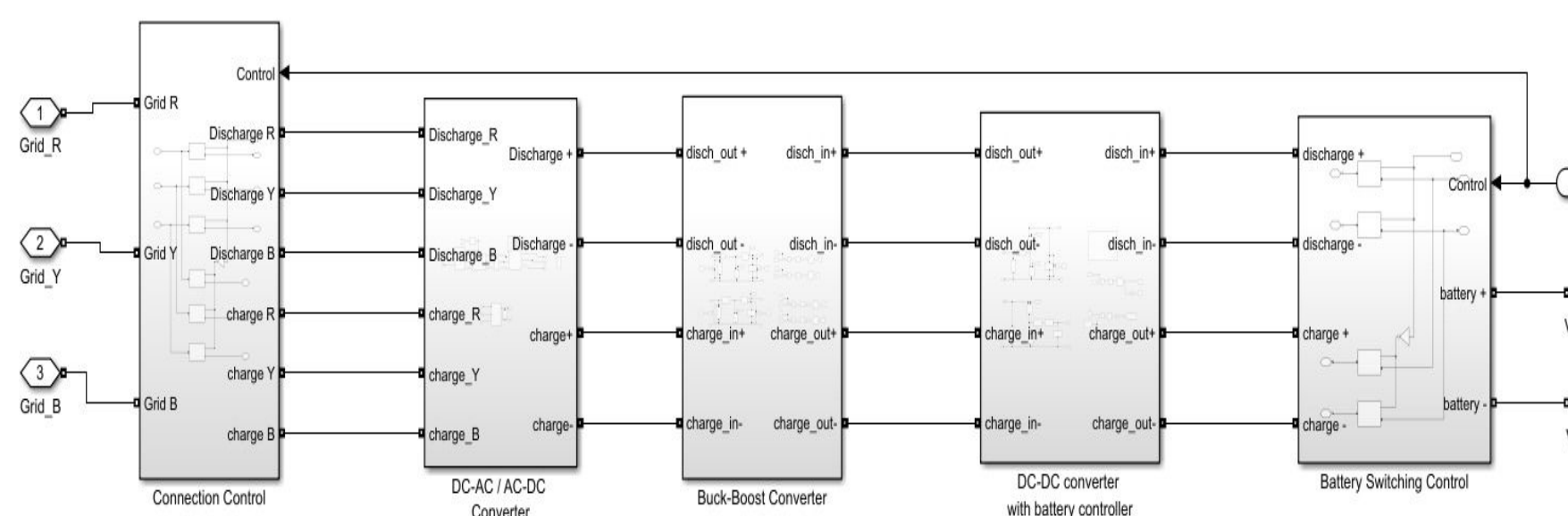


Fig. 3:Wall charger box.

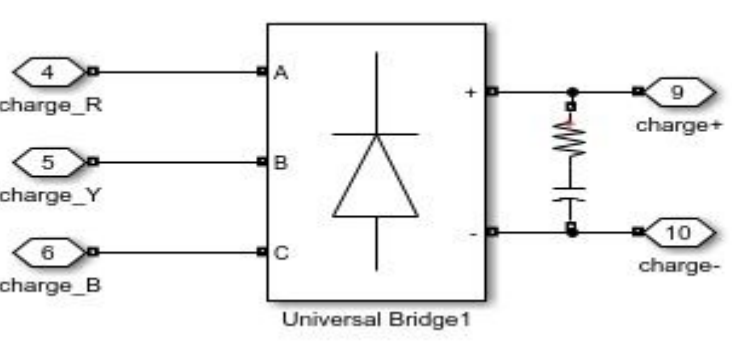


Fig. 5:Rectifier for G2V.

Battery Model

- EV battery model provides:
 - SOC
 - battery voltage
 - battery current.

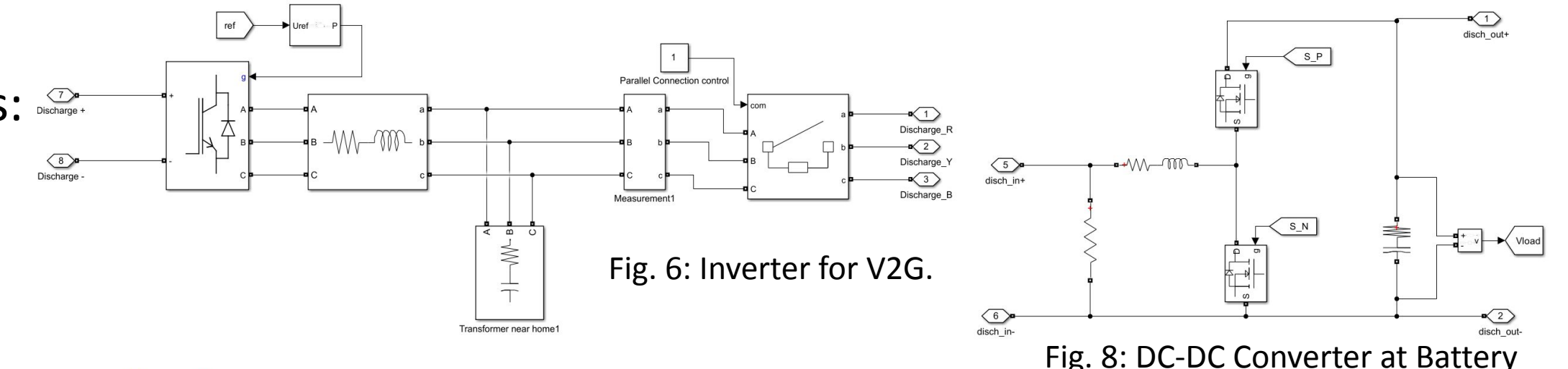


Fig. 6: Inverter for V2G.

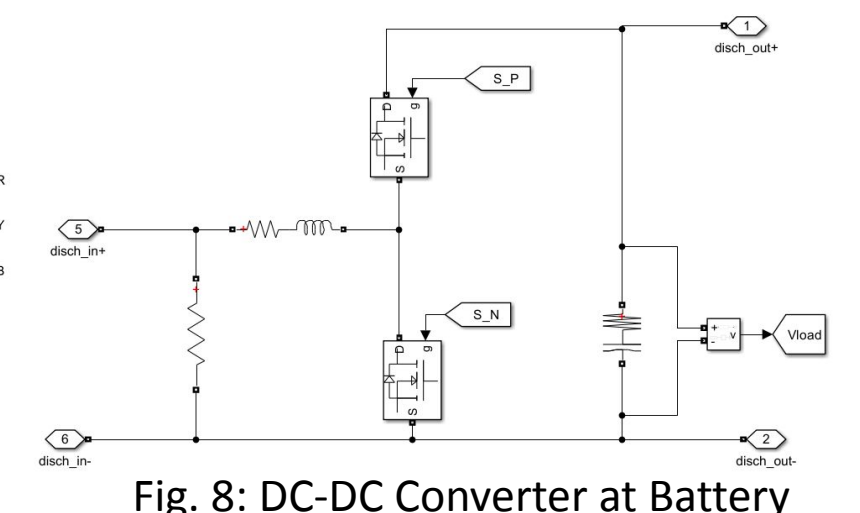


Fig. 8: DC-DC Converter at Battery

SOC estimation: $SOC(t) = SOC(0) - \frac{1}{Q_{bat}} \int I_b(t) dt$

Control Strategy (SOC + Price)

A simple rule-based control

is implemented using **SOC and electricity price**.

Control rules:

- If price is low and SOC < SOC max → charge battery (G2V)
 - If price is high and SOC > SOC min → discharge to grid (V2G)
 - Otherwise → idle (User Depend)
- $SOC_{min} = 20\%$
 - $SOC_{max} = 90\%$

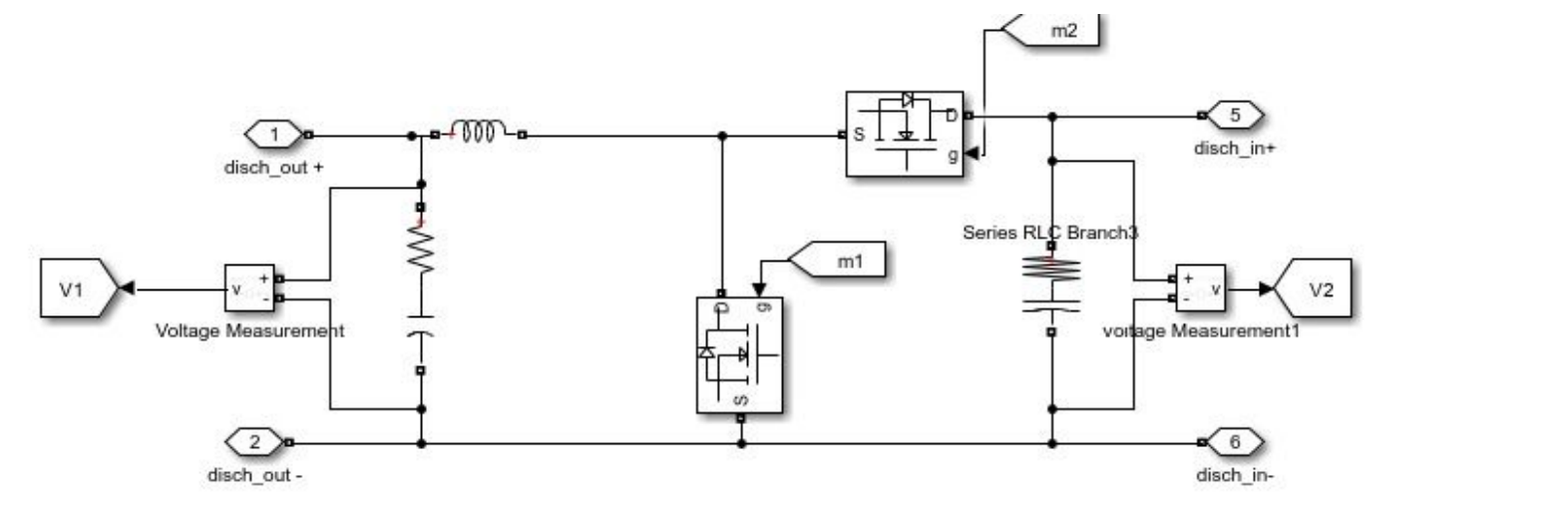


Fig.7: Bidirectional Buck-Boost Converter.

Initial Results

Simulation Results: Bidirectional EV Charging Behaviour

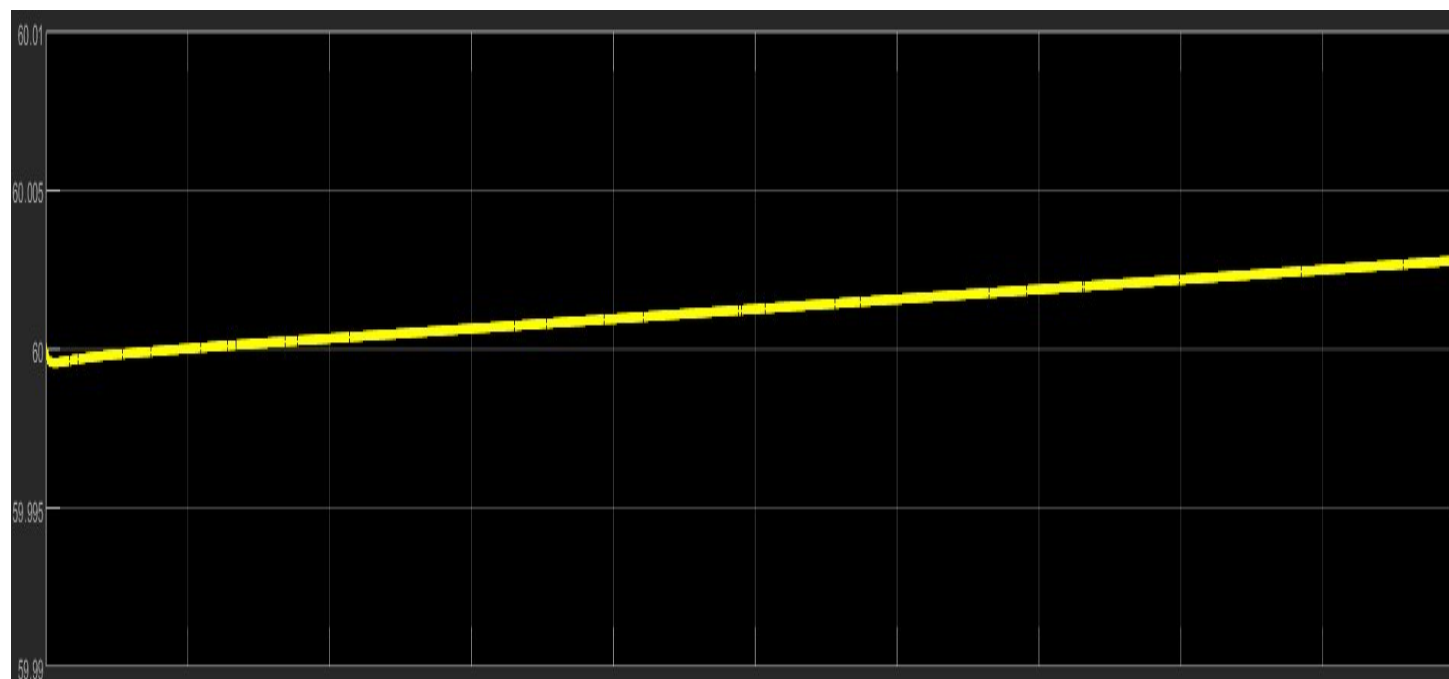


Fig. 9: SOC of battery at G2V.

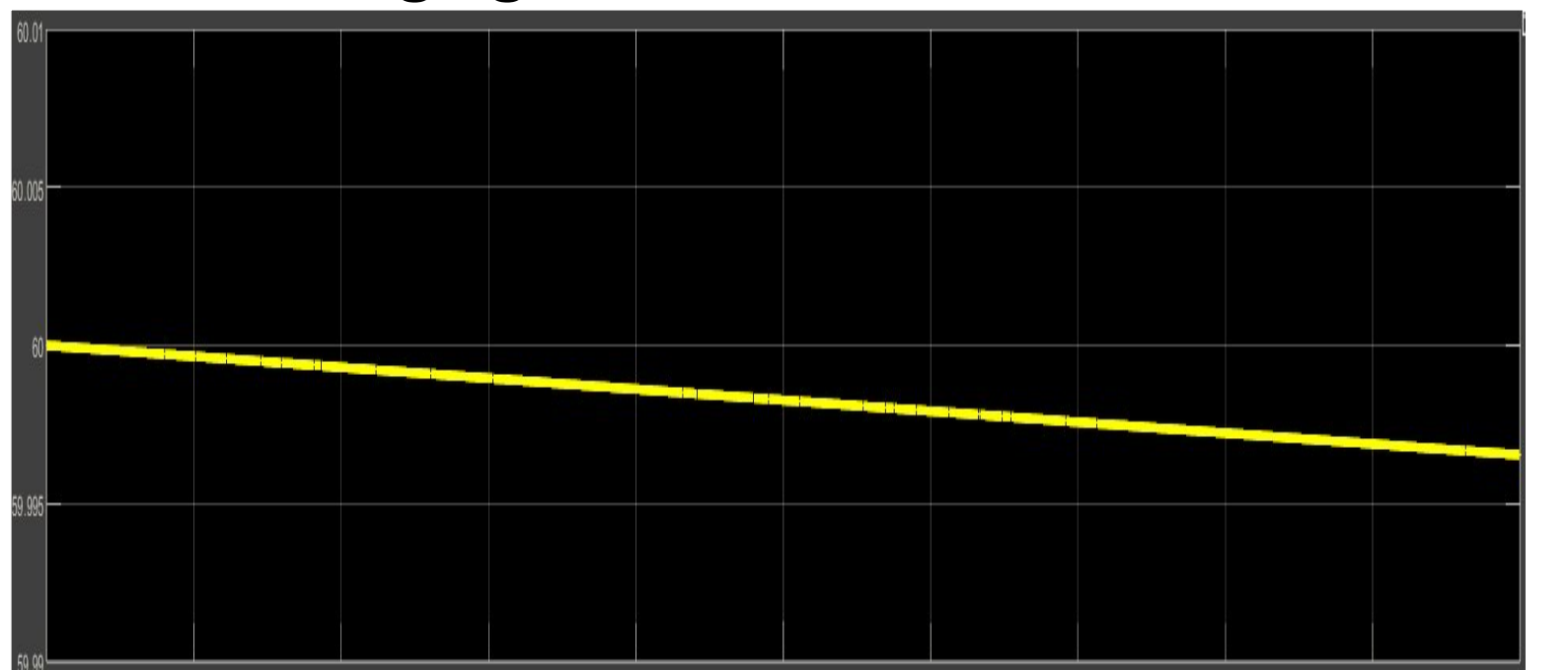


Fig. 12: SOC of battery at V2G.

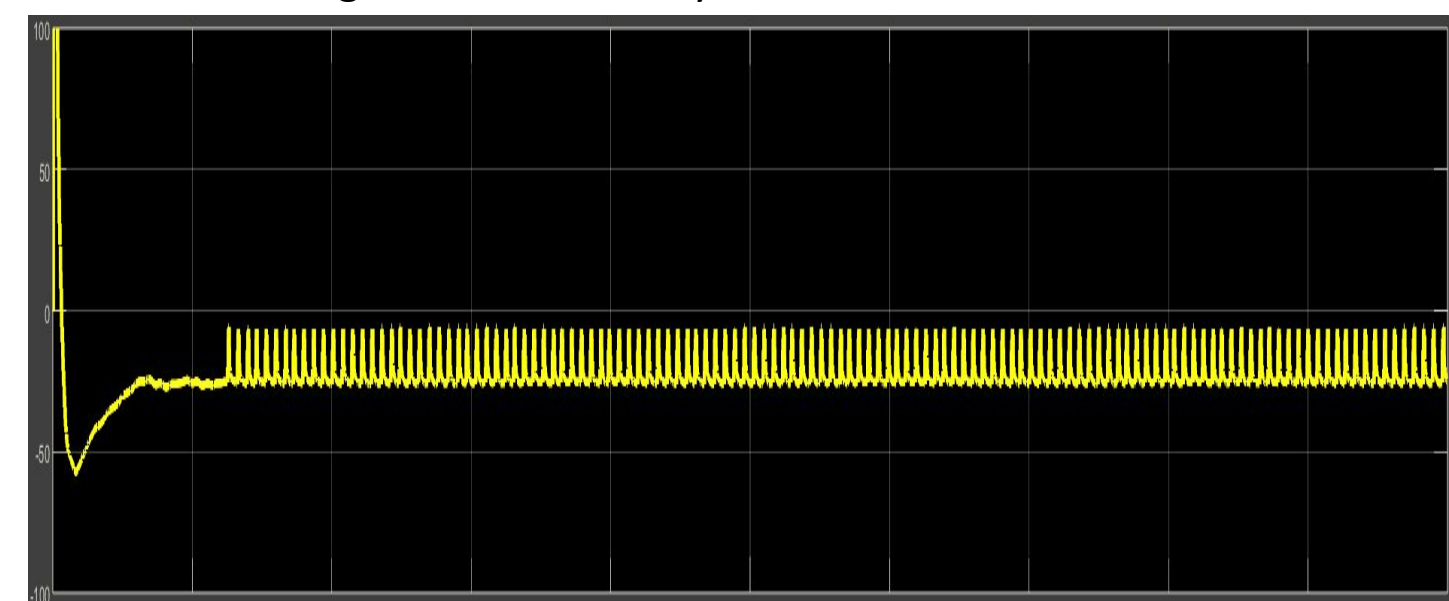


Fig. 10: Current of battery at G2V.

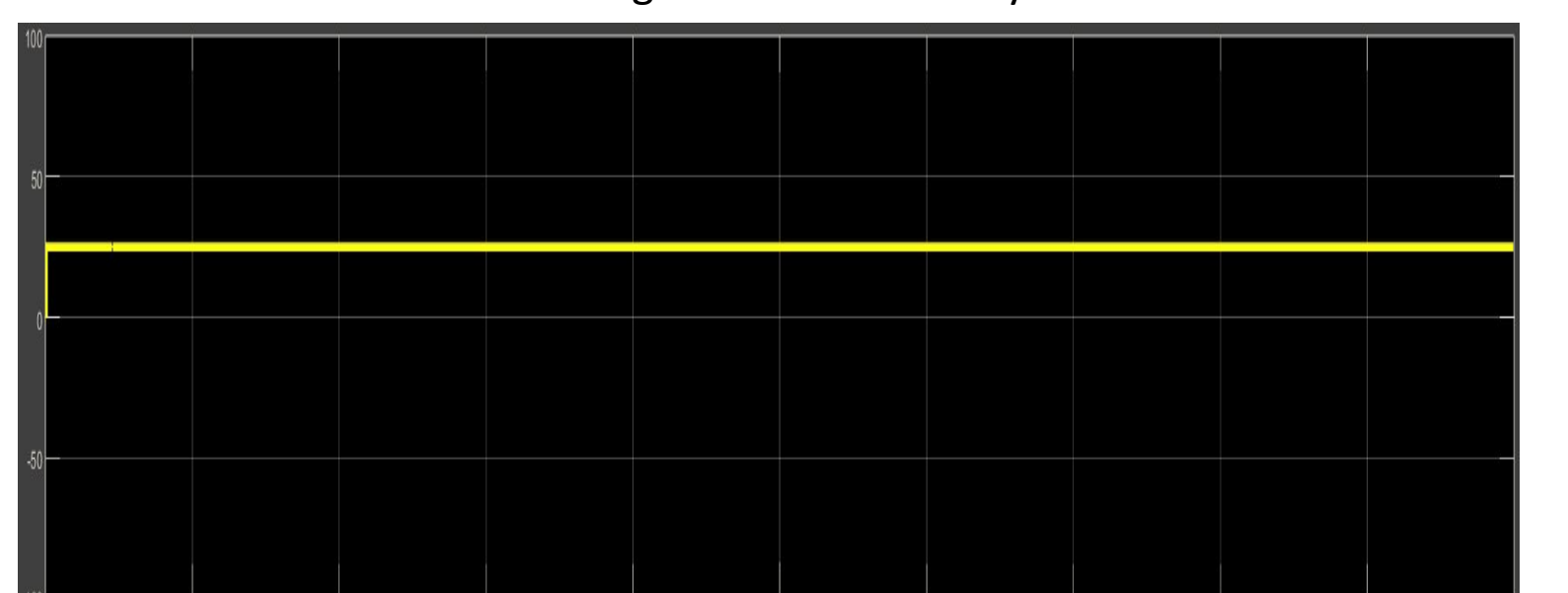


Fig. 13: Current of battery at V2G.

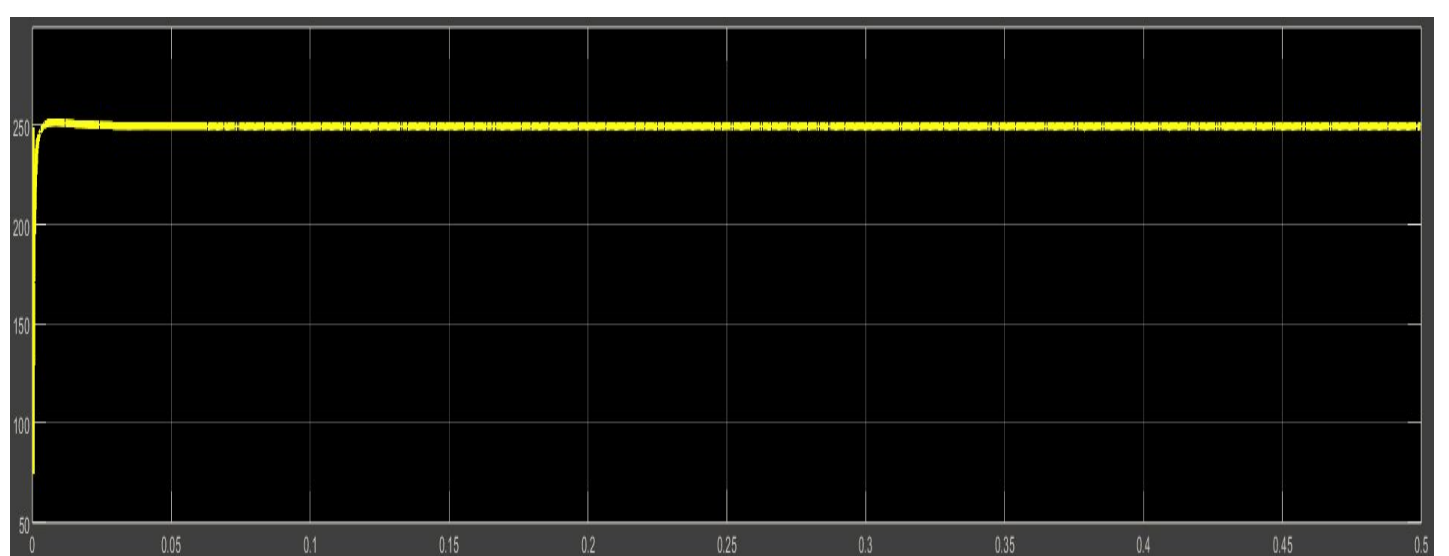


Fig. 11: voltage of battery at G2V.

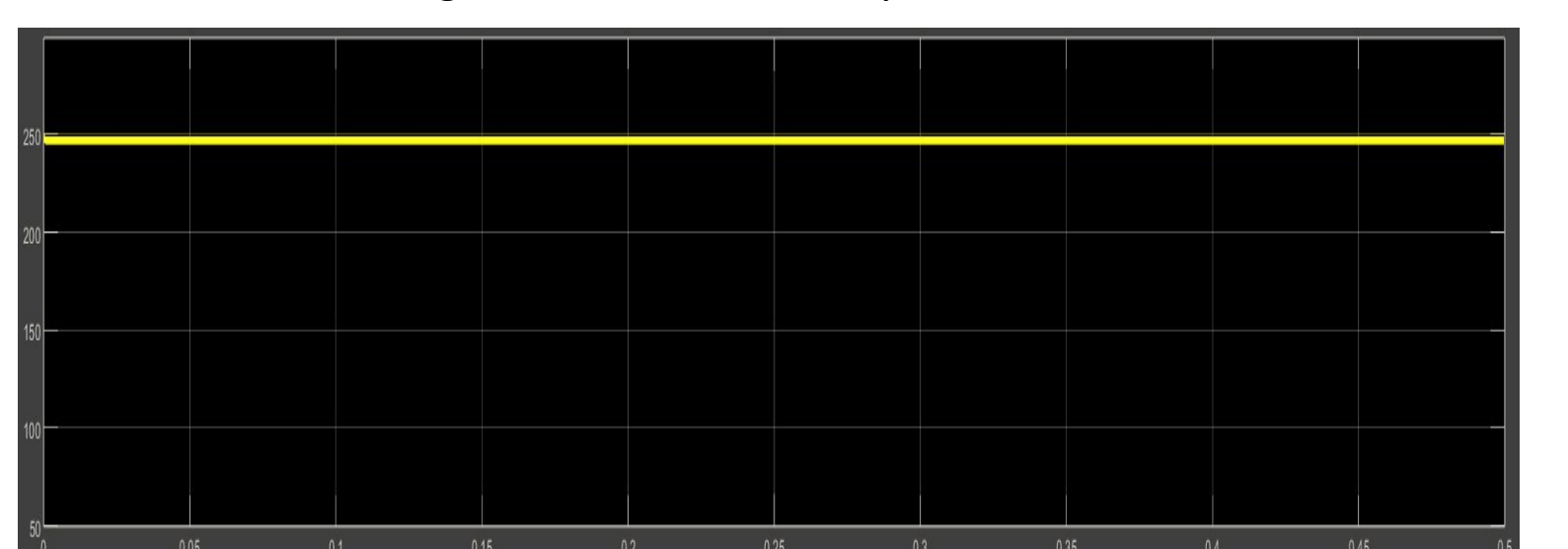
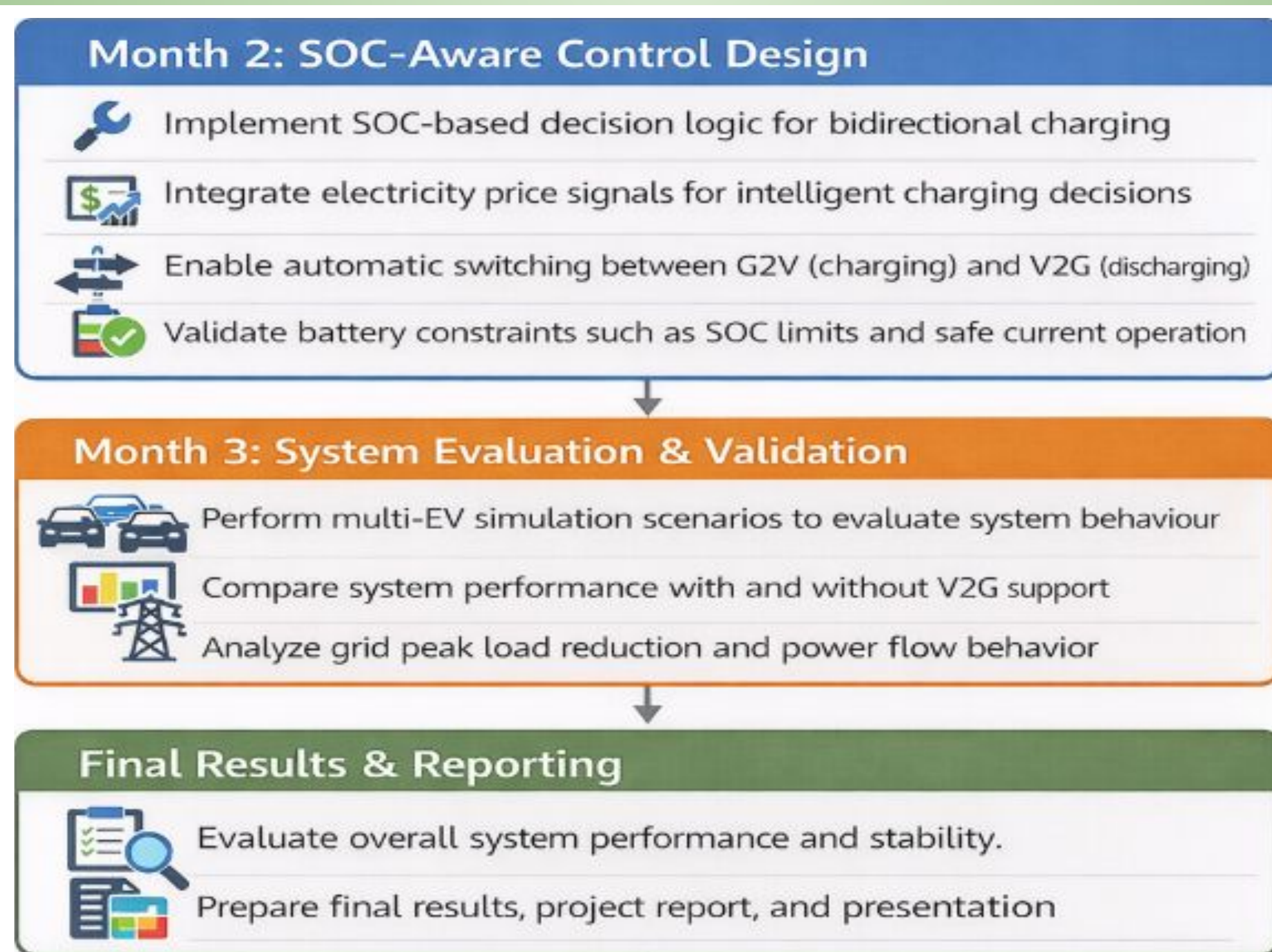


Fig. 14: Voltage of battery at V2G.

- In **G2V mode**, the EV battery is charged from the grid. The SOC increases gradually while the battery voltage rises towards its reference value. The charging current varies because the controller regulates the battery voltage using PID control.
- In **V2G mode**, the battery supplies power back to the grid. The SOC decreases slowly and the discharge current remains nearly constant once the system reaches steady operation, showing stable bidirectional power flow.

Plan Ahead: with milestones and timeline



References

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